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About Me:

I am a Computer Science student with a strong interest in backend development and full-stack web technologies. I have experience working with Flask, SQLAlchemy, and JavaScript, and I enjoy building structured, scalable web applications while improving my problem-solving skills.

AI/LLM usage

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, generate mock data and to get UI ideas. The extent of AI/LLM usage is around **10–15%**, limited to suggestions on UI and layout. All final Implementation logic, debugging, and integration were done manually.

Description

Project Title: Placement Portal Application

Problem Statement:

A **role-based web application** developed using **Flask** that streamlines campus placement activities by enabling structured interaction between **Admin (Institute Placement Cell), Companies,** and **Students**.

The application replaces manual processes such as spreadsheets and emails with a centralized system for managing company **approvals, placement drives, student applications, and placement history**.

Approach:

The app is built using **Flask** for its backend, has responsive UI build using **bootstrap**. It Allows role-based access to **Admin, Students and Companies** allowing the creation and management of drives and student applications. The admin controls access to students and companies.

Technologies used

Technology / Library

Flask

Purpose

Core backend web framework

SQLAlchemy

Object Relational Mapper for SQLite database

Jinja2

Template engine for rendering dynamic HTML pages

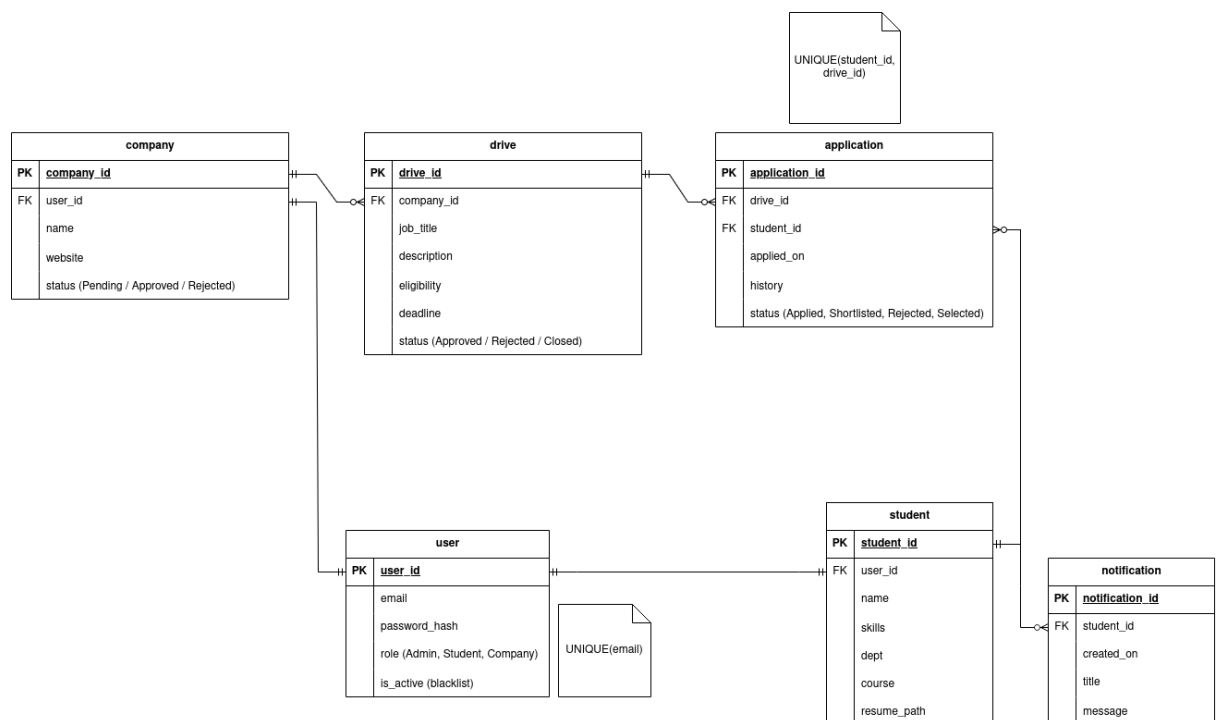
Bootstrap 5

Frontend styling and responsive design

SQLite

Lightweight local database for storing user data

DB Schema Design



Tables:

User — stores account details (user_id, email, password_hash, role, is_active)

Student — stores student profile (student_id, user_id, name, skills, dept, course, resume_path)

Company — stores company details (company_id, user_id, name, website, status)

Drive — stores placement drives (drive_id, company_id, job_title, description, eligibility, deadline, status)

Application — stores student applications (application_id, drive_id, student_id, applied_on, status, history)

Notification — stores messages for students (notification_id, student_id, created_on, title, message)

Relationships:

One-to-One → **User** → **Student**

One-to-One → **User** → **Company**

One-to-Many → **Company** → **Drive**

One-to-Many → **Drive** → **Application**

One-to-Many → **Student** → **Application**

One-to-Many → **Student** → **Notification**

Many-to-Many → Student ↔ Drive (via Application)

Constraint → UNIQUE(student_id, drive_id) prevents duplicate applications

API Design

No APIs were created, as it isn't a core requirement for the MAD 1 project.

Architecture and Features

- [app.py](#) - Flask application entry point
- **models.py** - database models using SQLAlchemy
- **/routes** - Flask Blueprints for auth, students, company and admin routes
- **/templates** - Jinja2 HTML templates
- **/static** - Custom CSS with icons form light / dark mode selector

Implemented Features:

- User registration and role-based login, with access control to restrict users from accessing other roles dashboards and endpoints.
- Student profile management including skills, department, course, and resume upload.
- Company registration with approval workflow (Pending / Approved / Rejected).
- Creation and management of placement drives by companies.
- Drive status handling (Approved / Rejected / Closed) with automatic deadline-based closure.
- Students can browse available drives and apply to eligible ones.
- Prevention of duplicate applications using unique constraints.
- Application tracking with current status (Applied, Shortlisted, Rejected, Selected).
- Application history tracking for status changes.
- Notification system to inform students about application status updates.
- Dashboard views for students and companies with relevant data summaries.

Video

<https://drive.google.com/file/d/18fG7ARh41e7ckAhPNwPyetS1mL6q56V8/view?usp=sharing>